

Maail Bista.

Batch : Jestha (Day)/Evening.

Statistics handwritten notes.

Statistics

Collect data $\longrightarrow$ Summarize data $\longrightarrow$ Analyze $\longrightarrow$ Make Conclusions

• Why Statistics in ML?

- understand data
- handle uncertainty
- find relationships
- test whether model B beats model A.
- make predictions and real-world decisions.

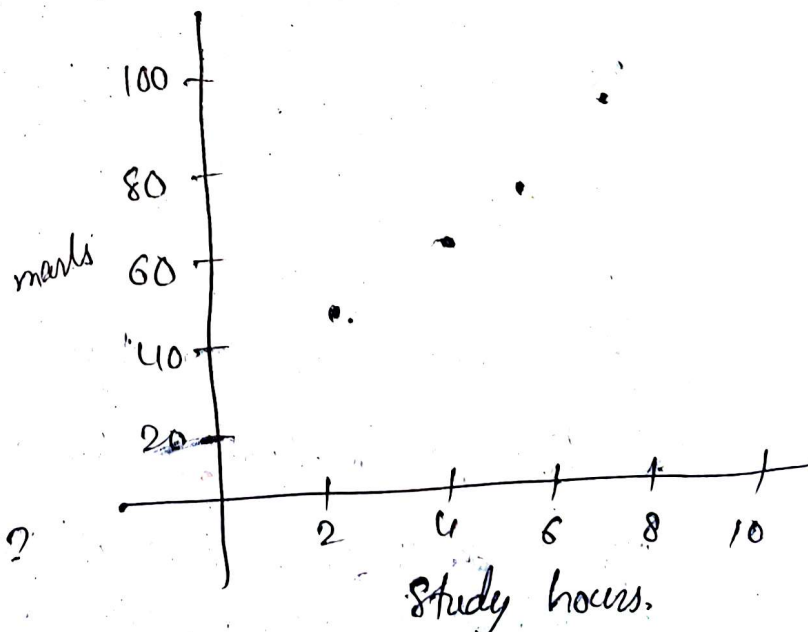
eg Student data

Student	Study hours	Attendance	Marks.
A	2	60%	45
B	4	70%	60
C	6	85%	75
D	8	95%	90

study hrs $\uparrow$ = marks $\uparrow$.

questions

- avg. marks?
- how much do marks vary?
- study more = higher score?
- attendance $\propto$ marks?
- from various attributes, can we estimate marks of another student?



Statistics



Descriptive

(summarize the data we have)
(mean, chart, table)



Inferential

(use a sample to conclude
something about a larger popⁿ)

Descriptive statistics

① Central tendency

the value around which data cluster around.

• mean

Data: 60, 70, 80, 90, 70.

$$\text{mean} = \frac{60+70+80+70+90}{5} = 74.$$

$$(\bar{x}) = \frac{\text{total of all values } (\sum x)}{\text{no. of values } (n)}.$$

or,

$$\bar{x} = \frac{\sum dx}{\sum f/n}$$

Used

- ① for data summarization (gives single value for large data)
- ② filling missing values (impute with mean).
- ③ Standardization (subtract the mean).
- ④ Group Comparison (eg, product A vs B).

⑩ Outlier detection (value far from mean).

• Median

1st sort in ascending / descending order.

find middle value.

Median = Value at $\left(\frac{n+1}{2}\right)^{\text{th}}$ term. (for odd no. of terms)

$$= \frac{\left(\text{Value at } \frac{n}{2} + \text{Value at } \frac{n}{2} + 1 \right)}{2} \quad (\text{for even no. of terms})$$

• Mode

• most frequent value from given data.

• most common occurrence.

Used,

① Survey results

② Categorical data analysis

③ Product or color popularity.

#2 Measure of Dispersion.

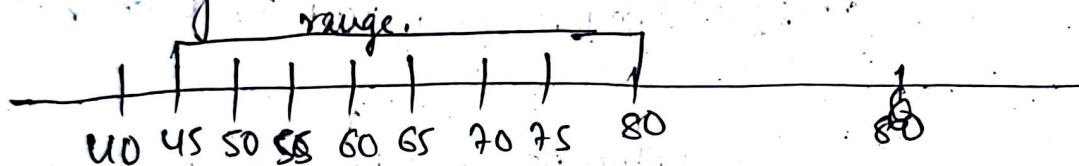
① Range

difference betⁿ max and min value of given datasets.

eg,

Score: 45, 50, 52, 58, 60, 67, 70, 80.

$$\text{Range} = 80 - 45 = 35.$$



larger range = more data spread. & vice versa.

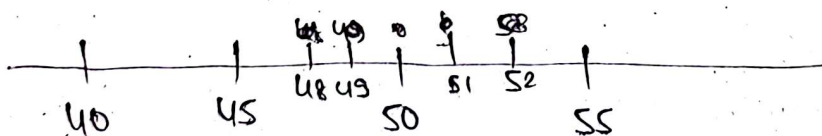
② Variance

how spread out values are around the mean.

②a low variance

Data: 48, 49, 50, 51, 52.

mean: 50.

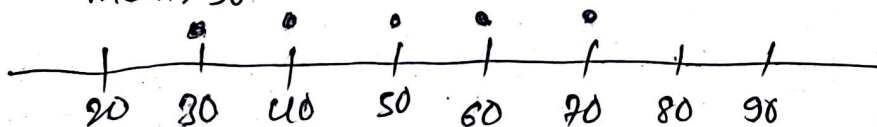


values stay close to the mean.

②b High variance

Data: 20, 40, 50, 60, 70.

Mean: 50.



values are spread far from the mean.

smaller variance = more consistency.

large variance = more variability.

⑩ Standard Deviation (σ)

- tells us how much the data values typically differ from the mean.
- measures the average distance of each value from the mean.
- small SD = values close to mean.

eg,

Student	Score (x)	$x - \mu$	$(x - \mu)^2$
A	70	$70 - 80 = -10$	100
B	75	$75 - 80 = -5$	25
C	80	$80 - 80 = 0$	0
D	85	$85 - 80 = 5$	25
E	90	$90 - 80 = 10$	100
Total	400	0	250

$$\mu = \frac{400}{5} = 80$$

$$\text{Variance } (\sigma^2) = \frac{250}{5} = 50$$

$$\text{SD } (\sigma) = \sqrt{\sigma^2} = \sqrt{50} = 7.07.$$

Why SD?

Variance gives the avg. of squared diff. from the mean, but it's unit is squared, which is hard to interpret.

But, $\sigma = \sqrt{\sigma^2}$, so same unit as the original data.
Variance

So, SD (σ) is easier to understand and interpret in real-world situations.

4. Percentile

p^{th} percentile means $p\%$ of the data is at or below that value.

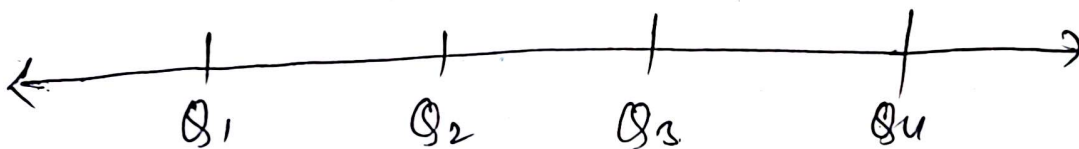
25th percentile (Q_1) = value at $\left(\frac{n+1}{4}\right)^{\text{th}}$ term

50th percentile (Q_2) = value at $\left(\frac{n+1}{2}\right)^{\text{th}}$ term

75th percentile (Q_3) = value at $\left(\frac{3(n+1)}{4}\right)^{\text{th}}$ term.

4th quantile
percentile (Q_4) = $\frac{4(n+1)}{4}^{\text{th}}$

= 100%. It data lies below this.



x	d	cd
10	2	2
20	4	6
30	6	12
40	8	17
50	3	20
	20	

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} \text{ term} = \left(\frac{21}{4}\right) = 5.25. \quad (\text{bet}^{\text{w}} 1^{\text{st}} \& 2^{\text{nd}} \text{ term}).$$

$$Q_1 = L + \frac{\left(\frac{n+1}{4}\right) - cd}{d} \times i$$

$$Q_2 = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term} = 10.5^{\text{th}} = (\text{bet}^{\text{w}} 2^{\text{nd}} \& 3^{\text{rd}} \text{ term})$$

$$Q_3 = \left\{\frac{3(n+1)}{4}\right\}^{\text{th}} \text{ term} = \left(\frac{63}{4}\right) = 15.75^{\text{th}} = (\text{bet}^{\text{w}} 3^{\text{rd}} \& 4^{\text{th}} \text{ term}).$$

Lower bound for Outliers.

Formula: $\boxed{\text{Lower Bound} = Q_1 - 1.5(IQR)}$

if, $Q_1 = 20$

$IQR = 10.$

then,
$$\begin{aligned} LB &= Q_1 - 1.5(IQR) \\ &= 20 - 15 \\ &= 5. \end{aligned}$$

Any value below 5 is considered potential outlier.

Upper bound

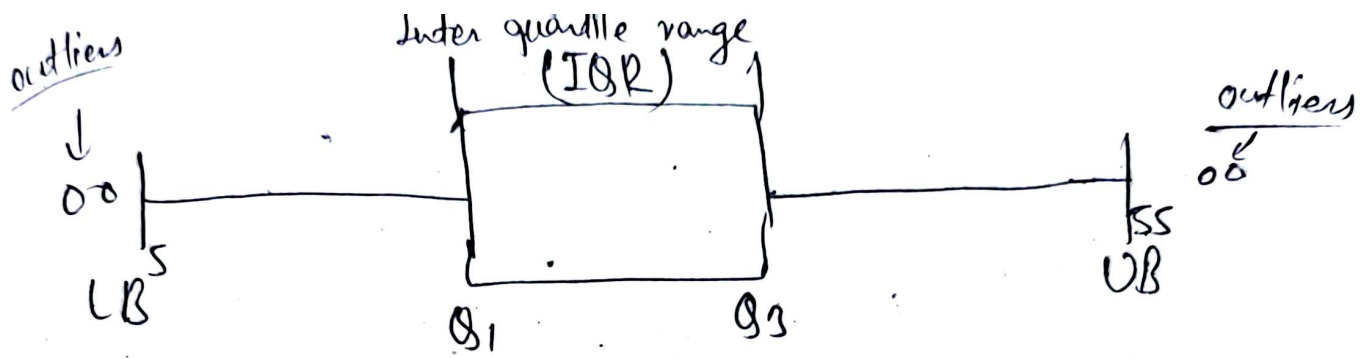
$$\boxed{\text{Upper bound} = Q_3 + 1.5(IQR)}$$

if, $Q_3 = 40$

$IQR = 10.$

then,
$$\begin{aligned} UB &= Q_3 + 1.5(IQR) \\ &= 40 + 15 \\ &= 55. \end{aligned}$$

Any value above 55 is considered a potential outlier.



$$IQR = Q_3 - Q_1.$$

if, $LB = -20$

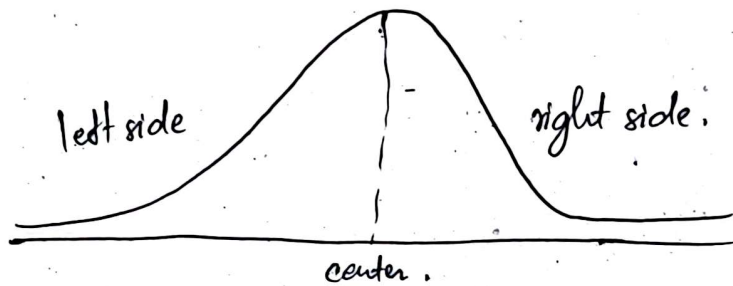
$UB = 44.$

$IQR = 16.$

Outlier = $\boxed{-20 \leq x < 44.}$

Symmetric Distribution

distribution where the data is spread almost equally on both sides of the center.



main property

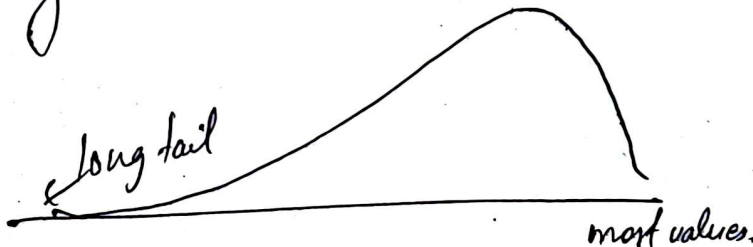
mean $\approx$ median $\approx$ mode.

In perfect symmetry,

mean = median = mode

① Left skewed Distribution.

most of the data are concentrated on right side; create a long tail on the left.



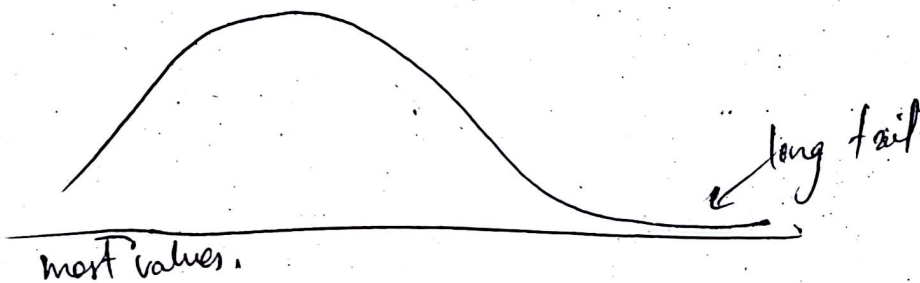
mean < median

eg, data : 20, 70, 75, 80, 85, 90, 95.
↓ small value. mostly high value

most value high, but a few low values stretch the tail left.

⑩ Right skewed

most values concentrated on left side leaving long tail on right.



mean > median

most values relatively low, few high values stretch long tail on right.

- # Covariance (tells the direction of relationship betⁿ 2 variables).
- Shows whether two variables move together in the same direction or in opposite direction.

Formula

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Range

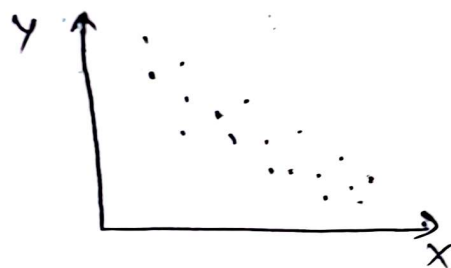
$$-\infty < \text{Cov}(X, Y) < \infty$$

if, $\text{Cov} > 0$: Positive covariance
(both variable tends to increase together)

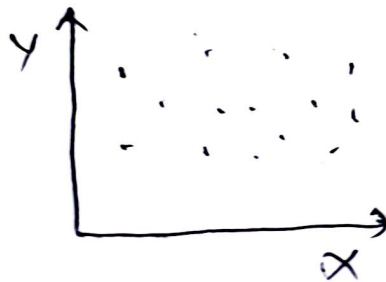


else if,

$\text{Cov} < 0$: Negative Covariance.
(both variable tends to decrease together).



$\text{Cov} \approx 0$: no clear linear relationship



- Covariance is not standardized, so its size alone does not directly tell the strength like correlation does.

Correlation

- measures the strength and direction of the linear relationship betⁿ two quantitative variables.
- tells us how variable changes w.r.t another.
- unitless (no unit).
- measures only linear relationship, not causation.
- based on covariance & SD (σ).

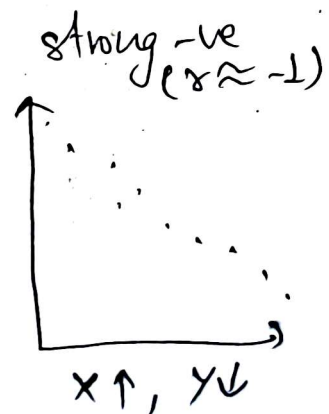
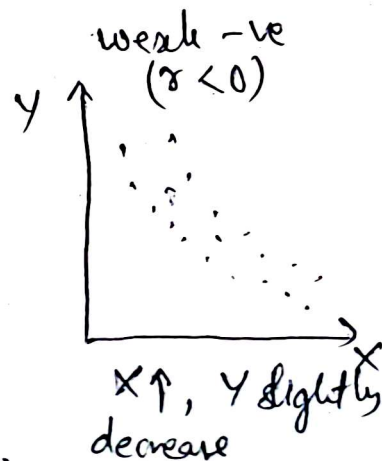
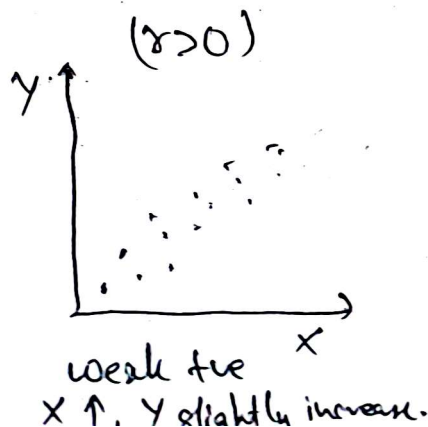
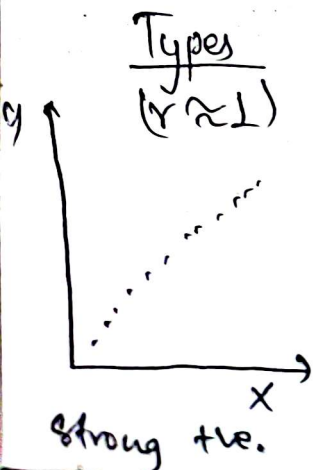
Ranges:

-1 to +1

+1 : perfect +ve linear relationship.

-1 : perfect -ve linear relationship.

0 : no linear relationship.



Formula (Pearson Correlation Coefficient)

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}, \quad -1 \leq r \leq 1.$$

where x = value of 1st variable
 y = value of 2nd variable.

eg,

$x : 1, 2, 3, 4, 5$

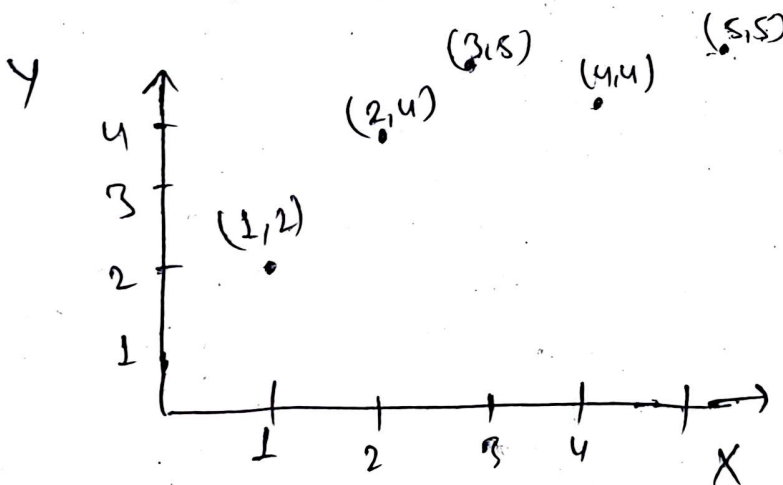
$$\bar{x} = 3$$

$y : 2, 4, 5, 4, 5$

$$\bar{y} = 4$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
1	2	-2	-2	4	4	4
2	4	-1	0	0	1	0
3	5	0	1	0	0	1
4	4	1	0	0	1	0
5	5	2	1	2	4	1
				6	10	6

$$r = \frac{6}{\sqrt{10} \sqrt{6}} \approx \frac{6}{7.746} \approx 0.775.$$



$0.775 \approx 1$, strong pos. correlation.

Z-Score and Standardization.

Z-score tells us how many SD(s) a value is above or below the mean.

Standardization converts a value into a Z-score.

Formula

$$Z = \frac{x - \mu}{\sigma}$$

x = actual value

μ = mean.

σ = standard deviation.

Z = Z-score

eg 1

$$x = 70$$

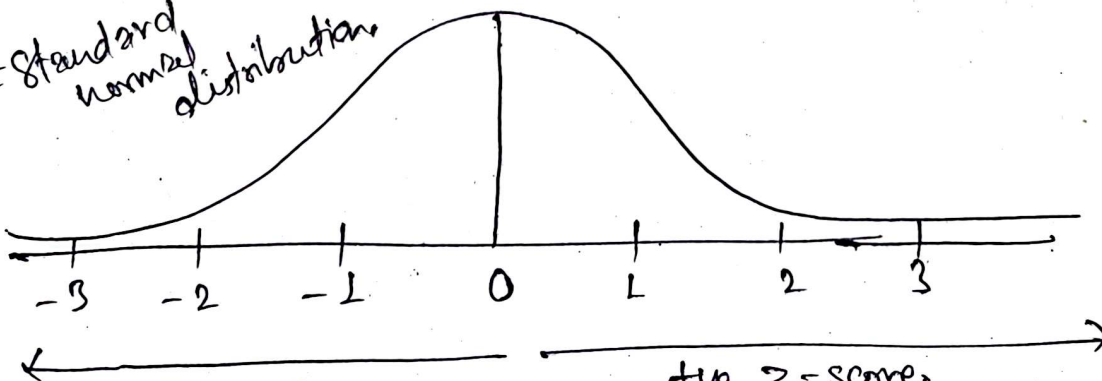
$$\mu = 50$$

$$\sigma = 10$$

$$Z = \frac{x - \mu}{\sigma} = \frac{70 - 50}{10} = 2$$

70 is 2 standard deviation above mean (50).

Standard normal distribution



larger $|Z|$ = farther from mean.

the Z-score.
(above the mean).

smaller $|Z|$ = closer to mean.

eg of Standardization.

① Before standardization,

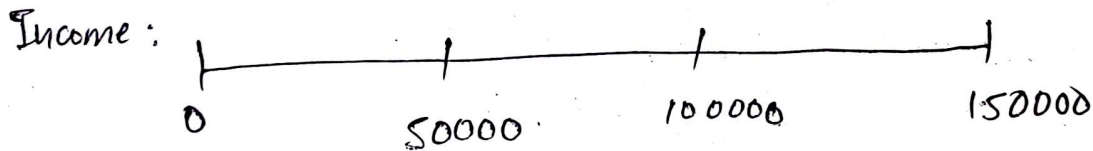
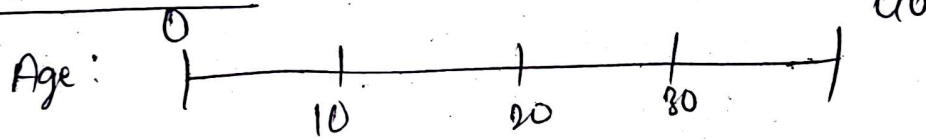
Sample	Age (years)	Income (USD)
1	18	20000
2	22	35000
3	25	50000
4	30	80000
5	35	120000

Feature

Age value = small

Income = large.

Feature scales



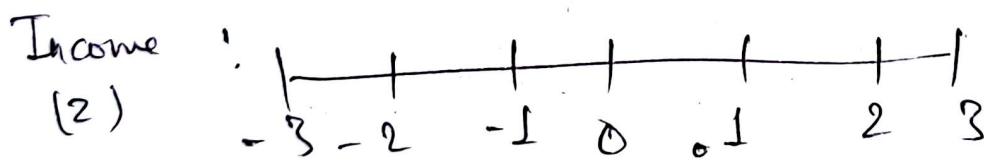
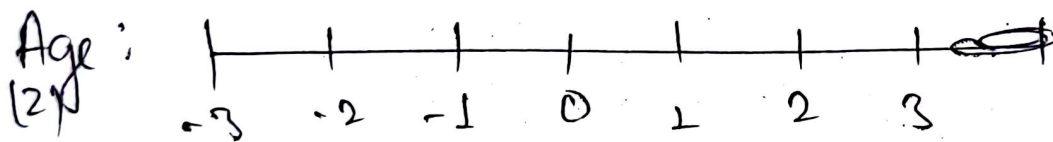
② Standardization (solution)

$$Z = \frac{x - \mu}{\sigma}$$

(11) After standardization.

Sample	Age (z)	Income (z)
1	-1.23	-1.08
2	-0.68	-0.69
3	-0.20	-0.29
4	0.61	0.50
5	1.50	1.50

now, both features are on similar scale.

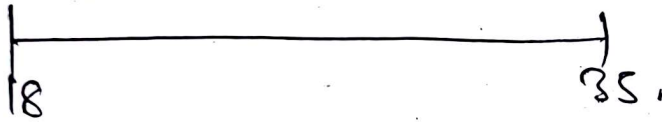


Similarly normalization

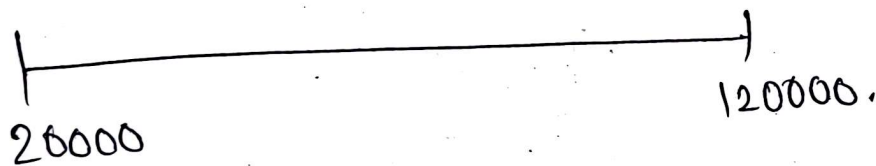
① Before

Sample	Age	Income
1	18	20000
2	22	35000
3	25	50000
4	30	80000
5	35	120000

Age (years)



Income (USD)



Problem: features have diff. scales.

large-scale features can dominate small-scale feature.

② Normalization

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Age = 25,

$x_{\min} = 18$

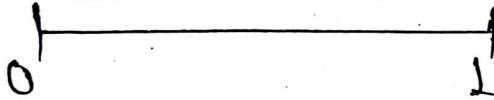
$x_{\max} = 35$

$$x' = \frac{7}{17} \approx 0.41, \text{ (bet}^n \text{ 0 to 1)}.$$

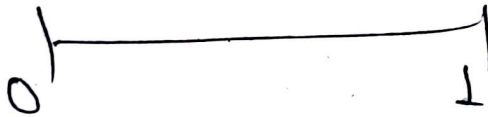
Adder normalization,

Sample	Age (normalized)	Income (normalized)
1	0.00	0.00
2	0.24	0.15
3	0.41	0.30
4	0.71	0.60
5	1.00	1.00

Age (normalized):



Income (normalized):



both feature lie on similar scale: 0 to 1.

Normalization

- Use normalization when you want a fixed range like 0 to 1.

Often used in KNN or k-means

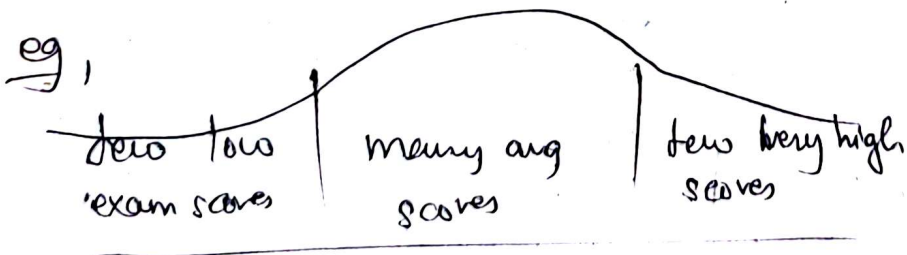
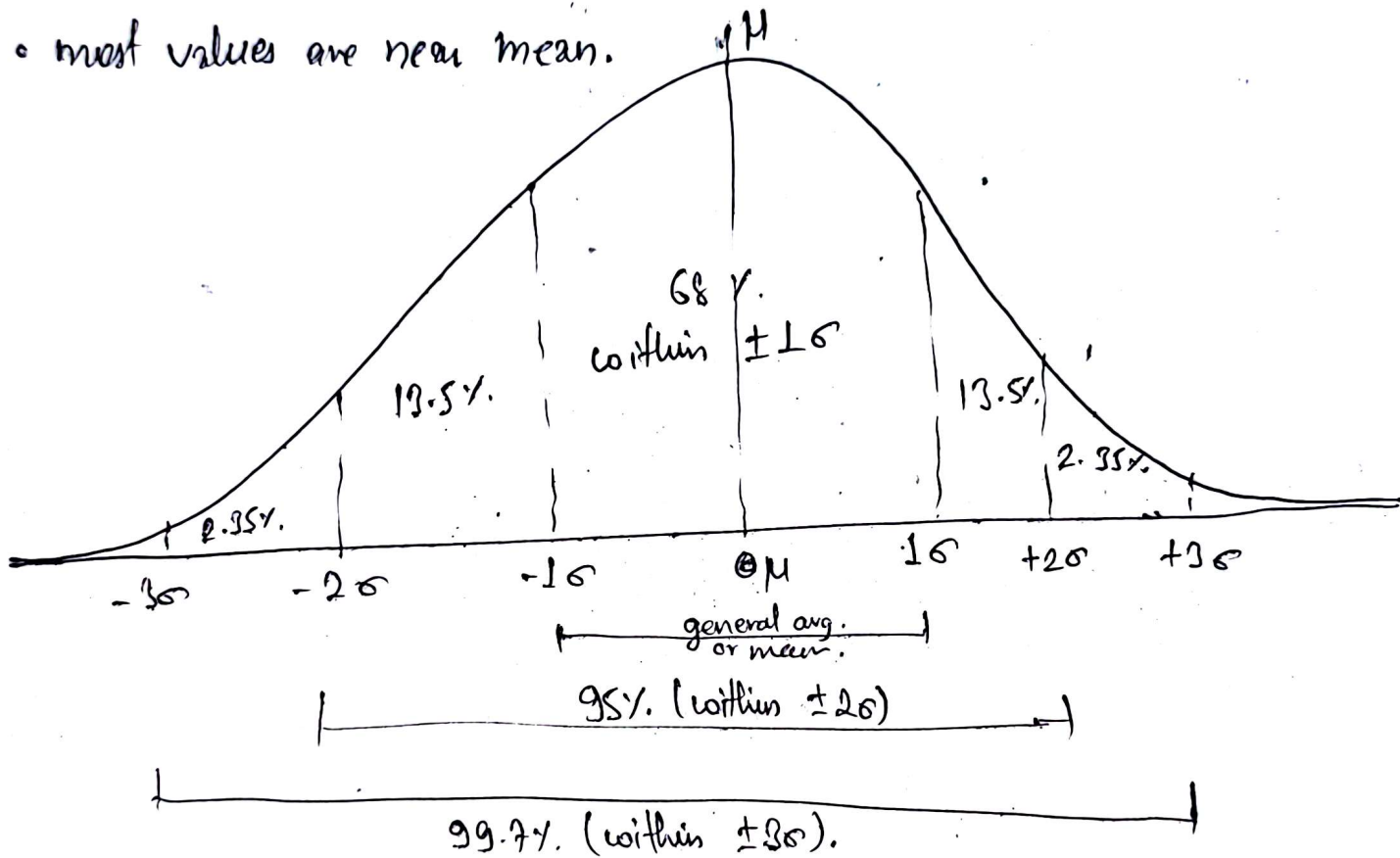
Standardization

- when features centered at 0 with standard deviation 1.

Common for linear Regression, SVM, logistic Regression.

Normal distribution

- Bell shaped
- Symmetric
- mean = median = mode.
- most values are near mean.



- 50% of data lies on each side of the mean.
- fewer value as we move away from the mean.
- Defined by mean (μ) and standard deviation (σ).

① μ (mu) = mean - center of distribution.

② $\mu \pm 1\sigma$: about 68% of values lie here.

1 SD $\pm$ from mean.

③ $\mu \pm 2\sigma$: about 95% value

④ $\mu \pm 3\sigma$: about 99.7% value

⑤ Beyond $\mu \pm 3\sigma$: very rare values and are often considered outliers.

Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad X \sim N(\mu, \sigma^2)$$

x = data value

μ = mean

σ = SD

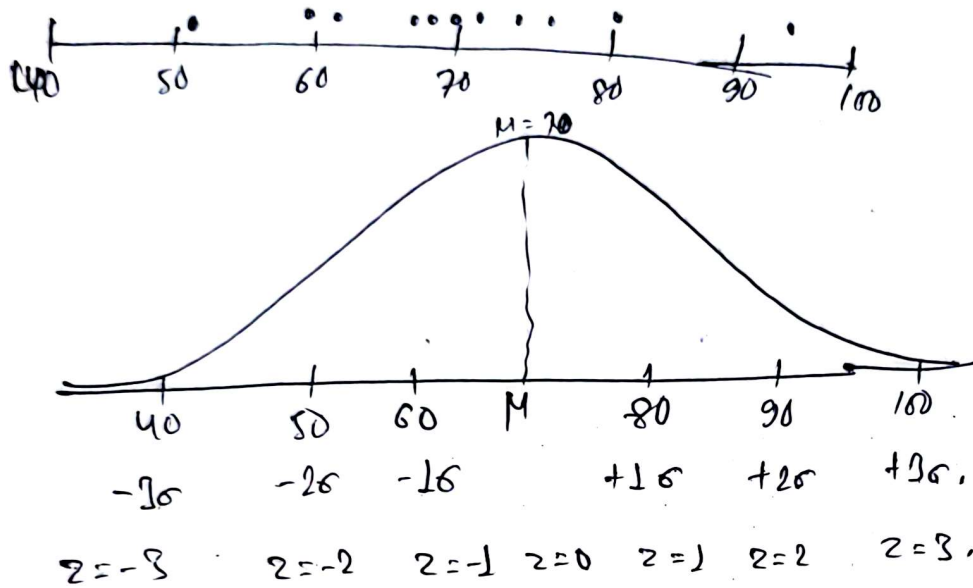
σ^2 = variance

$\pi \approx 3.1416$

$e \approx 2.718$.

① ^{eg}, let, $\mu = 70$
 $\sigma = 10$.

② Distribution plot



③ let, $x = 85$.

$$z = \frac{x - \mu}{\sigma} = 1.5$$

Score 85 is 1.5 SD above the mean.

Used

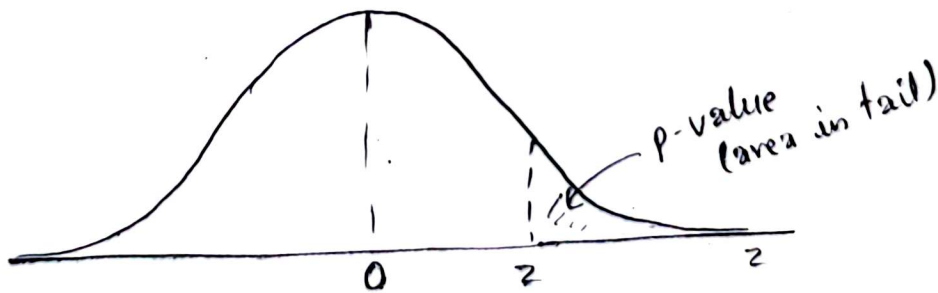
- ① In many distance based ML algo. (KNN, k-means)
- ② Linear models (Linear / Logistic Regression).
- ③ SVM (performs better with scaled, roughly normal features).
- ④ Neural networks (training is more stable with normalized data).

Assumptions for hypothesis testing

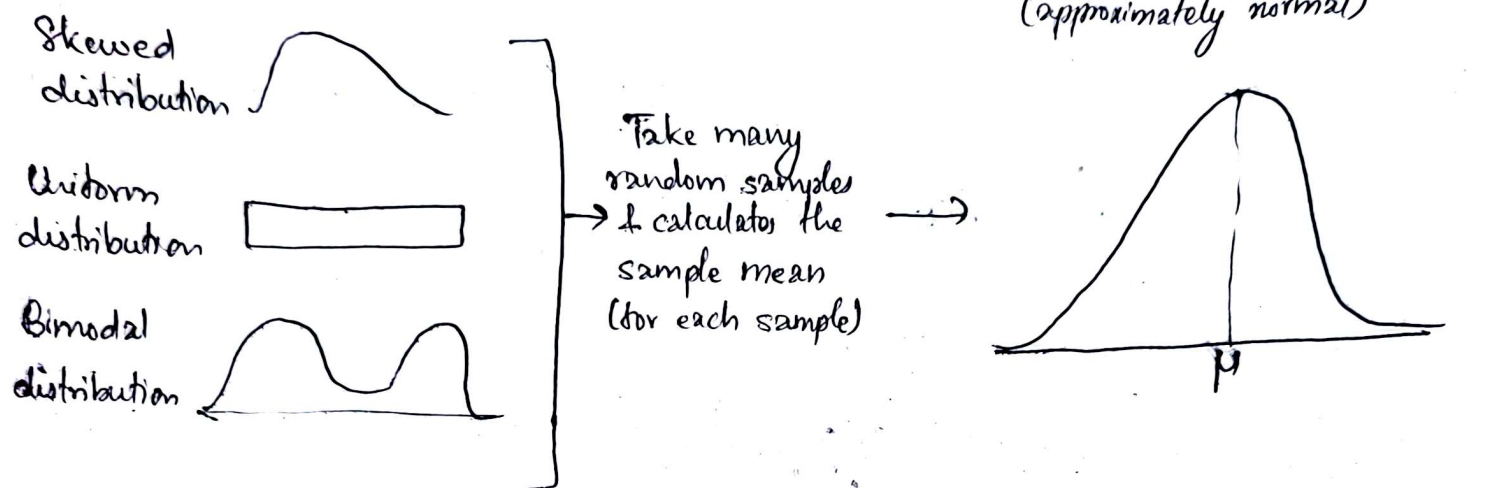
- many statistical tests assume that data (or residuals) are approximately normally distributed, especially for small samples.
 - t-test (comparing means)
 - ANOVA (comparing multiple groups)
 - Regression (assumes normally distributed residuals)
 - Confidence intervals
 - many parametric tests.

P-values and Statistical Inference

normal distribution helps us calculate probabilities, p-values, and confidence intervals.



Central Limit Theorem.

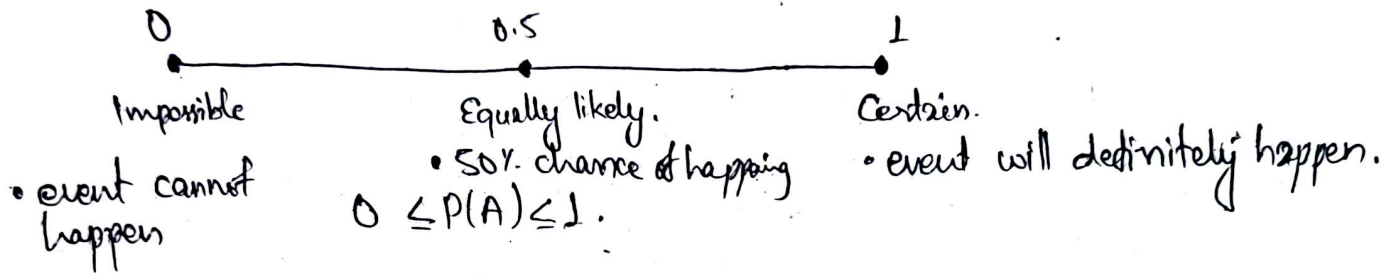


Useful in Other Areas

- Modeling measurement errors (often assumed normal)
- Quality control and process monitoring.
- In finance
- In natural and social sciences.

Probability

measures how likely an event is to happen.



Experiment = an action with uncertain result.

Outcome = one possible result.

Sample Space (S) = all possible outcomes.

Event (A) = a set of outcomes we care about.

eg roll a dice.

Sample Space = $\{1, 2, 3, 4, 5, 6\}$

Event (A) = Getting an even no. = $\{2, 4, 6\}$

$$P(A) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

$$P(\text{even}) = \frac{3}{6} = 0.5$$

Probability eg,

Drawing an Ace from 52 cards.

$$\text{total ace} = 4$$

$$\text{total card} = 52.$$

$$P(\text{ace}) = \frac{4}{52} = \frac{1}{13} \approx 0.0769 \approx 7.69\%.$$

$$P_{\text{not}}(\text{ace}) = \frac{48}{52} = \frac{12}{13} \approx 0.9231 \approx 92.31\%.$$

Rule of Probability

① Probability Range

$$0 \leq P(A) \leq 1.$$

Probability is always betⁿ 0 & 1.

② Complement Rule

$$P(A^c) = 1 - P(A).$$

Probability of not A = 1 - Probability of A.

③ Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A & B are mutually exclusive;

$$P(A \cup B) = P(A) + P(B).$$

A or B

④ Multiplication Rule

$$P(A \cap B) = P(A) \cdot P(B|A)$$

A & B

If A & B are independent:

$$P(A \cap B) = P(A) P(B).$$

⑤ Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given that B has happened:

⑥ Total Probability

$$P(A) + P(A^c) = 1.$$

An event and its complement add to 1.

Additional rule

① Mutually Exclusive Example.

A = rolling 1

B = rolling 6.

cannot happen together.

$$\begin{aligned}P(1 \text{ or } 6) &= P(1) + P(6) \\&= \frac{1}{6} + \frac{1}{6} \\&= \frac{1}{3}.\end{aligned}$$

Multiplication rule

If A & B are independent, then $P(A \cap B) = P(A) \times P(B)$.

① Independent Events

Prob. of getting 6 on 1st roll & 6 on 2nd.

$$P_1(6) = \frac{1}{6}$$

$$P_2(6) = \frac{1}{6}.$$

$$P(6 \& 6) = \frac{1}{6} * \frac{1}{6} = \frac{1}{36}.$$

1st roll doesn't affect 2nd roll.

⑪ Dependent Events

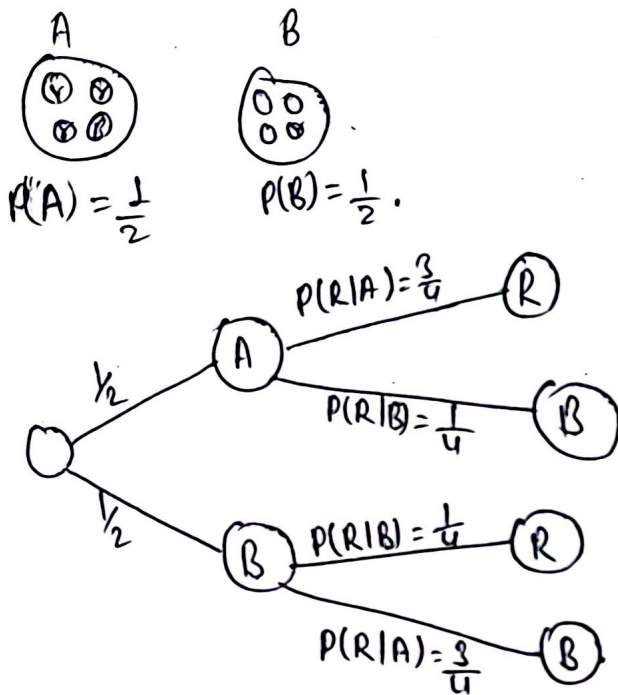
Eg; Probability of getting ace as 1st card and ace as 2nd.

$$P(1^{\text{st}} \text{ ace}) = \frac{4}{52}$$

$$P(2^{\text{nd}} \text{ ace}) = \frac{3}{51}$$

$$P(\text{Ace \& Ace}) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$

Bayes' theorem



$$P(R) = P(R|A) P(A) + P(R|B) P(B)$$

$$= \frac{3}{4} \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{2}$$

$$= \frac{3}{8} + \frac{1}{8}$$

$$= \frac{1}{2}$$

$$= 0.5.$$

$$P(A|R) = \frac{P(R|A) \times P(A)}{P(R)}$$

$$= \frac{\frac{3}{4} \times \frac{1}{2}}{\frac{1}{2}}$$

$$= \frac{3}{4}$$

$$= 66.67\%$$

Q. mobile shop gets its phones from 3 companies: A, B & C.

Company A

Supplies 50% of phone.

2% of A's phone are defective.

Company B

Supplies 30%.

defective = 3%.

Company C

Supplies = 20%.

defective = 5%.

Solⁿ

$$P(D) = P(A) P(D|A) + P(B) P(D|B) + P(C) P(D|C)$$

$$= 0.5 \times 0.02 + 0.3 \times 0.03 + 0.2 \times 0.05$$

$$= 0.029.$$

$$P(C|D) = \frac{P(D|C) \times P(C)}{P(D)} = \frac{0.05 \times 0.2}{0.029} = 0.3448.$$

Probability that defective phone came from company C = 0.3448

$$\approx 34.48\%$$

Random variable

Discrete
Random variable

Continuous Random
Variable

eg.

Coin toss = 3 times

let X = no. of heads.

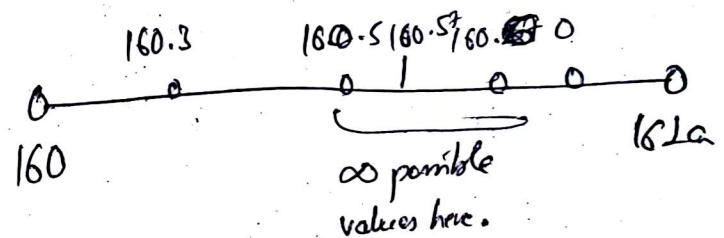
Outcomes	X
T T T	0
H T T / T H T / T T H	1
H H T / H T H / T H H	2
H H H	3

$$X = \{0, 1, 2, 3\}$$

• takes countable values.

• takes any value in a range.

X = height of a student.



Probability distribution

- shows what values a random variable can take and how likely each value is.

eg,

Coin toss = 2 times.

let X = no. of heads.

Possible values: $\{0, 1, 2\}$.

List all possible outcomes:

$$S = \{HH, HT, TH, TT\}$$

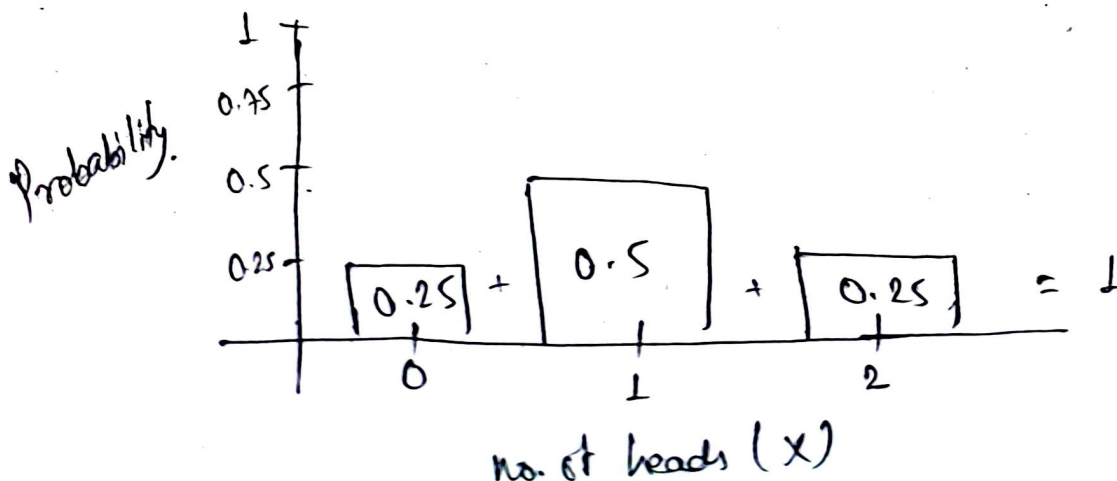
(each outcome is equally likely, with probability $\frac{1}{4}$).

Find the probability for each value of X :

$$P(X=0) = P(TT) = \frac{1}{4} \cdot (0.25)$$

$$P(X=1) = P(HT \text{ or } TH) = \frac{2}{4} \cdot (0.5)$$

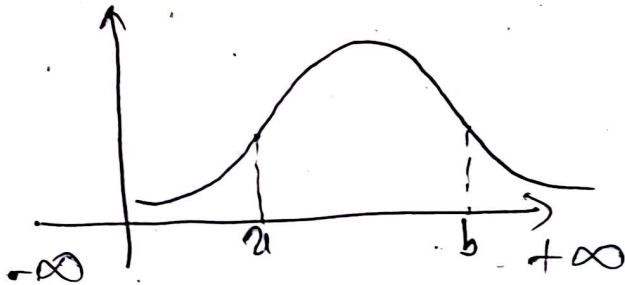
$$P(X=2) = P(HH) = \frac{1}{4} \cdot (0.25).$$



"When random variables (x) are represented with their probabilities $f(x)$ or $P(x)$, then this arrangement is called probability distribution".

Continuous random variables	x	$f(x)$	} probability.
	x_1	$f(x_1)$	
	x_2	$f(x_2)$	
	x_n	$f(x_n)$	

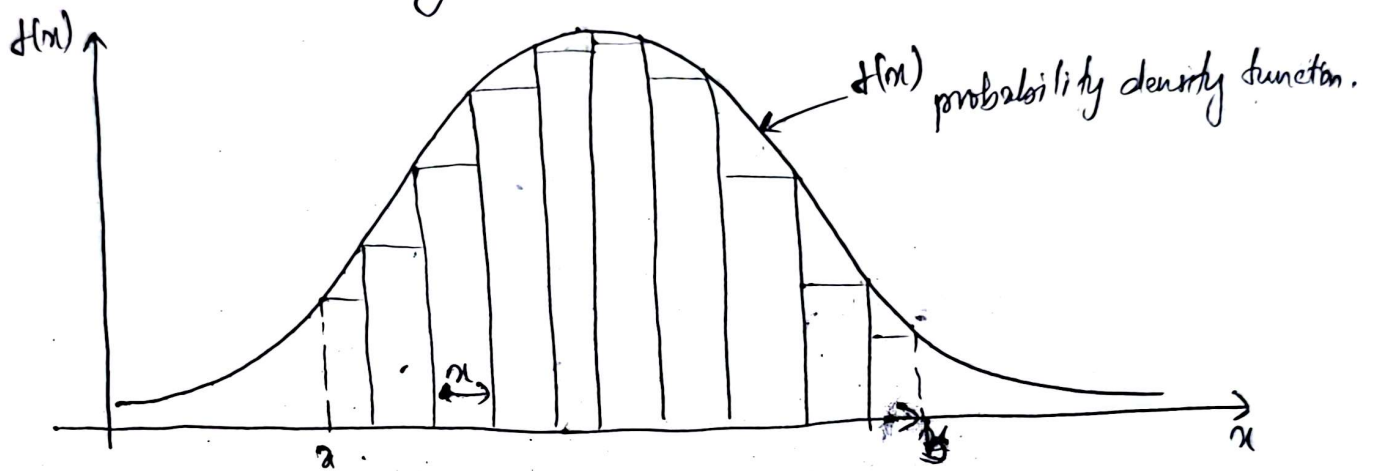
"If random variables are continuous then this arrangement is called probability density function".



$P(a \leq X \leq b) = \text{Area under curve.}$

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

Continuous Probability Distribution.



Each strip (rectangle) has width Δx and height $f(x_i)$.

$$P(a \leq x \leq b) \approx \sum_{i=1}^n f(x_i) \Delta x$$

In the limit (as the number of strips becomes very large), this becomes an integral.

$$P(a \leq x \leq b) = \int_a^b f(x) dx.$$

Probability Mass Function (PMF)

PMF tells the probability of each possible value of a discrete random variable.

Formula

$$p(x) = P(X=x).$$

Used for:

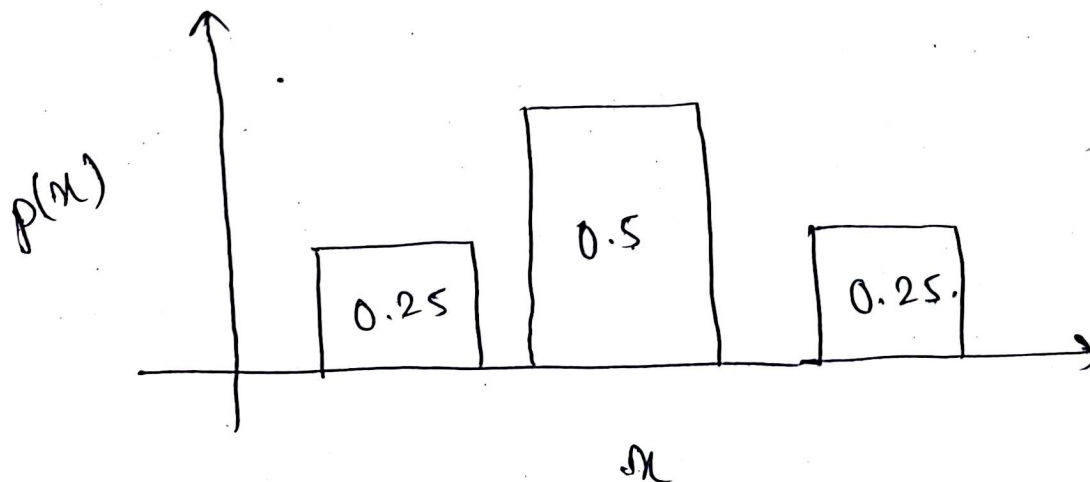
- Discrete random variables.

Countable values: $0, 1, 2, 3, \dots$

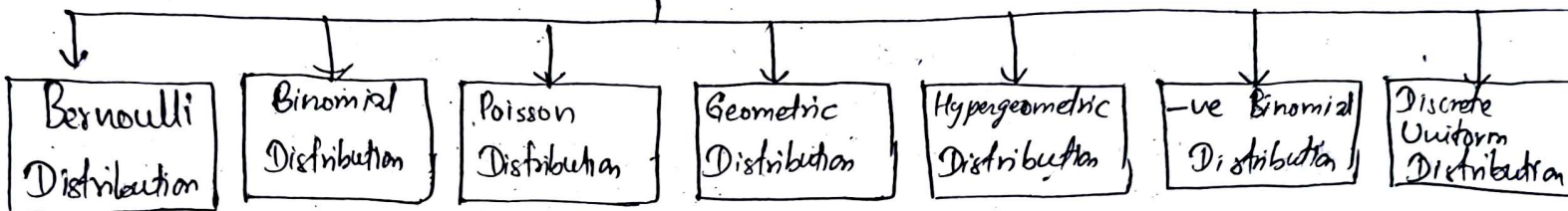
eg, let $X =$ no. of heads in 2 coin tosses

Outcomes	X	
TT	0	$\frac{1}{4} = 0.25$
HT	1	$\frac{2}{4} = 0.5$
TH	1	
HH	2	$\frac{1}{4} = 0.25$

1.



Discrete Probability Distribution



Binomial Distribution.

2 possible outcomes in each trial.

Success / failure.

Fixed no. of trials

T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇	T ₈	T ₉	T ₁₀
----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------	-----------------

$n = 10$ fixed trials.

Binomial distribution: repeated independent trials, two outcomes each trial, fixed no. of trials.

Formula

$$P(X = x) = {}^n C_x p^x q^{n-x}$$

n = no. of trials

x = no. of success.

p = prob. of success. ($0 \leq p \leq 1$)

q = prob. of failure ($q = 1 - p$).

Conditions

- (i) Fixed no. of trials (n)
- (ii) 2 outcomes/Trial (success or failure)
- (iii) Trials are independent
- (iv) Probability of success (p) is the same for every trial.

eg,

Toss a fair coin 3 times

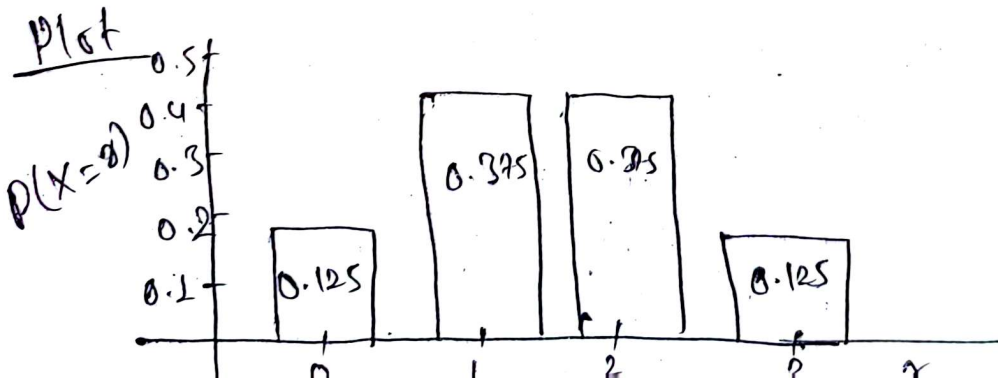
X = no. of heads

here, $n=3$ & $p=0.5$ ($q=0.5$).

Probability Distribution

x	$P(X=x)$
0	0.125
1	0.375
2	0.375
3	0.125

$$P(X=2) = {}^3C_2 (0.5)^2 \cdot (0.5)^{3-2} \\ = 0.375.$$



Binomial Distribution in ML & Data Science.

① Binary Classification

Yes/No outcomes such as spam vs not spam.

② A/B testing

no. of users who convert out of n visitors

③ Healthcare prediction

Patient +ve vs -ve for a condition.

④ Quality Control

Defective vs non defective products.

⑤ Marketing Analytics

Ad click or no click, purchase vs no purchase.

⑥ Customer Analytics

Churn vs no churn, response vs no response.

Poisson Distribution

gives the probability of a certain no. of events occurring in a fixed interval of time, space, distance, or any other interval, when the avg. rate of occurrence is constant.

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

X = no. of events

λ = avg. no of events in the interval ($\lambda \geq 0$)

$e \approx 2.718$

$x!$ = factorial of x .

when it is used

- ① when counting how many times an event happens in a fixed interval
- ② events occur independently
- ③ avg. rate of occurrence is constant in the interval.

eg, website receives on average 3 request/min.
So, $\lambda = 3$.

Find prob. of getting exactly 2 req. in 1 min.

Solⁿ

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \frac{e^{-3} 3^2}{2!}$$

$$= \frac{e^{-3} \times 9}{2}$$

$$\approx 0.224.$$

$$\approx 22.4\%.$$

Meaning

$P(X=2)$: means the probability that exactly 2 events occur in given interval.

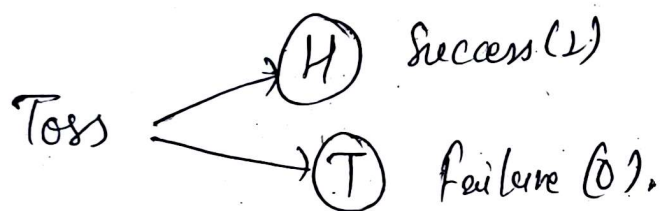
P = probability

X = no. of events

2 = exactly 2 events

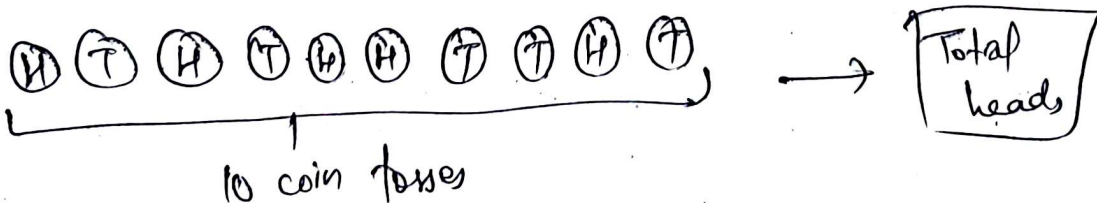
Bernoulli Distribution.

- only one trial
- Two possible outcomes only.
- Success (1) or failure (0).
- eg, Toss a coin once $\rightarrow$ Head or tail



Binomial Distribution.

- multiple repeated Bernoulli trials.
- each trial has two outcomes
- counts no. of successes
- eg, Toss a coin 10 times $\rightarrow$ no. of heads



Sampling and Hypothesis testing

Population: entire group we want to study. (r

Sample: smaller group taken from the population.
(when too costly to analyze entire popⁿ).

Parameter vs Statistics

Parameter

- A numerical value that describes the whole population.

Population: 10,000 students

True avg. height: $\mu = 165 \text{ cm}$.

165 cm is a parameter.

Population $\longrightarrow$ Parameter.

Statistic

- a numerical value calculated from a sample

Sample : 200 randomly selected students.

Sample avg. height : $\bar{n} = 164.2 \text{ cm.}$

164.2 is static.

Sample $\longrightarrow$ Statistic.

Use a sample statistic to estimate a popⁿ parameter.

#

Sampling Bias

Random Sampling

- Student selected randomly.
- $n = 200$

(Random sample)

Representative
 $\rightarrow$ Low Bias.

Convenience Sampling

- only student in classroom are selected

$n = 200$

(Classroom stds)

Biased
not representative

Voluntary Response Sampling

- only stds who choose to respond. (online survey)

$n = 200$

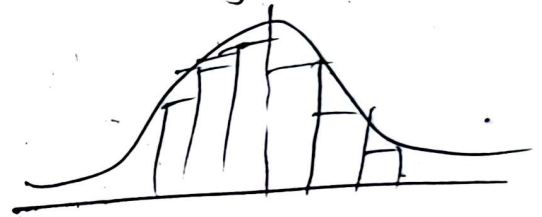
(Student who responded)

Biased
not representative

Sampling Distribution (of the Sample mean)

Population $\longrightarrow$ Take many random samples $\longrightarrow$ Calculate sample mean for each sample

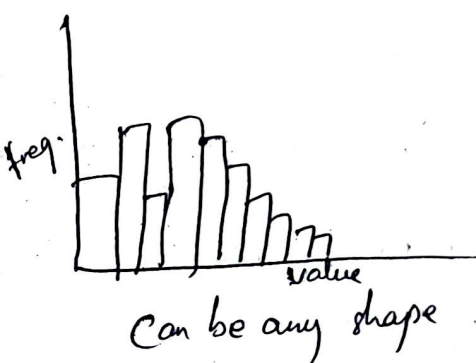
Sampling distribution



Central Limit Theorem.

If we take many large random samples and compute the mean of each sample, the sample means are approximately normal.

1. Population



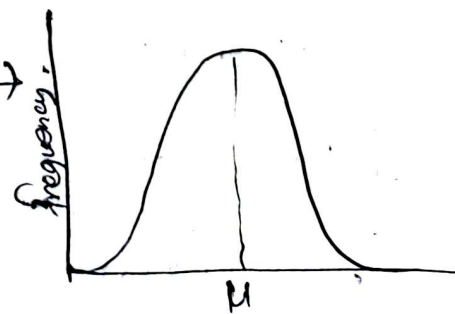
2. Many random samples

Sample 1 $\rightarrow$ mean = 62

Sample 2 $\rightarrow$ mean = 66

Sample 3 $\rightarrow$ mean = 64

3. Sample means



Sample mean ($\bar{x}$)
Approx. normal.

Hypothesis Testing

Statistical method used to decide whether a claim about a population is likely true or not, using sample data.



- used to check whether a claim is supported by sample data and used to decide if a result is real or just due to chance.

Why used?

- To test a claim about a population.
- To decide if sample results are meaningful
- To compare groups, methods or models.
- To check whether a difference happened by chance.

Benefits

- Gives obj. decisions
- Reduces guesswork.
- Supports research and business decisions.
- Shows statistical significance clearly.

#

Null hypothesis (H_0)

- the starting assumption.
- usually means no effect, no diff, or no change.

eg,

H_0 : new teaching method doesn't improve marks.

Alternative hypothesis (H_1)

- what we want to check.
- there is an effect, a difference, or a change.

eg,

H_1 : new teaching method improves marks.

Why useful in Data Analysis & ML?

In Data Analysis

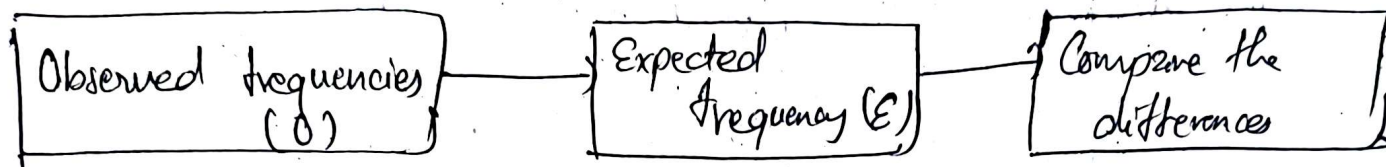
- Test whether sample results are meaningful.
- Checks relationships betⁿ variables.
- Compare two groups or methods.
- Support evidence-based conclusions.

In machine learning

- Checks if a new model is really better.
- Compare model A vs model B fairly
- Test whether a feature improves performance.
- Reduce decisions based only on guesswork.

Chi-square Test

hypothesis testing method mainly used with categorical data.

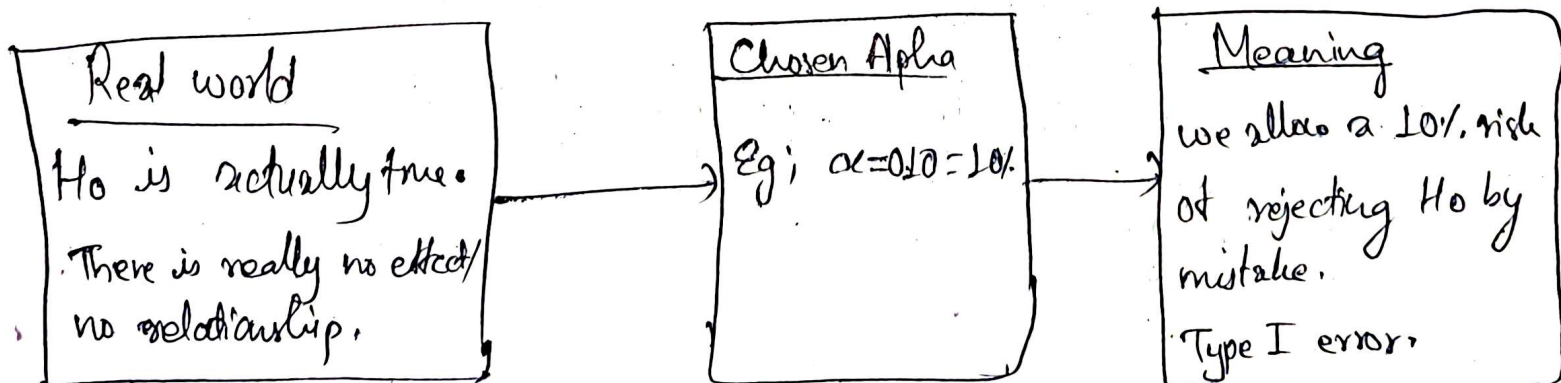


Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

What does Alpha (α) mean?

- allowed risk of rejecting (H_0) when H_0 is actually true in the real world.



Higher $\alpha \rightarrow$ easier to reject H_0 , but higher risk of being wrong.

Degree of freedom

$$df = (r-1)(c-1)$$

no. of independent cell values that can vary.

eg) 2×2 table.

$$df = (2-1)(2-1) = 1$$

	A	B	Row total
Row 1	a	60 - a	60
Row 2	50 - a	a - 10	40
Column total	50	50	100

4 cells, but only 1 can be chosen freely.

So, $df = 1$ means only 1 independent cell value can vary.

When is Hypothesis testing useful in Data Science?

- comparing two models or two groups.
- checking whether a change improved results
- validating a claim from sample data.

eg,

A company tests whether a new recommendation model increases click-through rate.

H_0 : the new model does not improve click-through rate.

H_1 : the new model improves click-through rate.

Hypothesis testing helps decide whether the improvement is real enough to trust.